KSP Datathon: The Ultimate Action Blueprint

Your Best Friend

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Part 1: Required Software Apps to Install

Before writing a single line of code, you must download and install these desktop applications on your laptop:[cite: 1]

- **Visual Studio Code (VS Code):** The code editor where you will type all your Python scripts.[cite: 1]
- **Docker Desktop:** Required to package your app for Zoho.[cite: 1] It must be installed and running in the background while you code.[cite: 1]
- **DB Browser for SQLite:** A visual tool to open and inspect the fake cybercrime database you generate.[cite: 1]
- **Postman:** An app to test the Bhashini Translation API and local LLM connections before adding them to your code.[cite: 1]
- **Node.js:** Required only because the Zoho Catalyst Command Line Interface (CLI) needs it to run on your laptop.[cite: 1]

Part 2: Deep Step-by-Step Execution Plan

Step 1: Generate Synthetic Police FIR Data

Actionable Steps:[cite: 1]

1. Open VS Code and create a new folder called `KSP_Datathon`.[cite: 1]
2. Open the terminal in VS Code and install the required libraries by typing: `pip install faker pandas`.[cite: 1]
3. Copy the AI prompt below, paste it into an AI, and put the generated Python code into a file named `generate_data.py`.[cite: 1]
4. Run the script.[cite: 1] It will output multiple CSV files.[cite: 1]
5. Open DB Browser for SQLite, import your new CSV files, and visually verify that the 1.5 Lakh rows of FIR, Accused, Victim, and CrimeHead data look realistic according to the KSP schema.[cite: 1]

AI Prompt: Data Generation
"Act as a Python Data Engineer.[cite: 1] Write a script using the 'Faker' library and Pandas to generate 150,000 synthetic records strictly based on the Karnataka Police FIR System schema.[cite: 1] Create interconnected relational datasets for 'CaseMaster', 'Victim', 'Accused', 'Complainant Details', and 'Arrest Surrender'.[cite: 1] Ensure the 'CrimeNo' in 'CaseMaster' perfectly follows this format: 1 digit Case Category Code + 4 digit District ID + 4 digit Police Station ID + 4 digit Year + 5 digit Running Serial Number.[cite: 1] Use lookup tables for 'CrimeHead', 'CrimeSubHead', and 'Gravity Offence' and maintain proper Foreign Key relationships.[cite: 1] Save outputs as interconnected CSV files."[cite: 1]

Step 2: Build the Agentic AI Core

Actionable Steps:[cite: 1]

1. Download and install Ollama, then run `ollama pull llama3` in your terminal to get the local model.
2. In your VS Code terminal, install the AI frameworks: `pip install langchain langchain-ollama sqlalchemy`.
3. Use the AI prompt below to get the LangChain code.[cite: 1] Paste it into `ai_agent.py`.[cite: 1]
4. Run the code in VS Code and type a test question in the terminal (e.g., "Find the CrimeNo linked to the highest number of accused persons") to verify the AI searches your relational CSVs.[cite: 1]

AI Prompt: Agentic Core
"Act as an AI Architect.[cite: 1] Write a Python script using LangChain's 'create_sql_agent'. Connect it to a local SQLite database containing the exact Karnataka Police FIR System schema data.[cite: 1] Use a local open-source LLM via Ollama (`langchain-ollama`). Ensure the agent converts a natural language query into SQL, executes it across the relational tables (such as CaseMaster and Accused), and returns a logical summary."[cite: 1]

Step 3: Integrate Kannada Voice & Text

Actionable Steps:[cite: 1]

1. Go to the Bhashini portal (or use OpenAI Whisper) and generate an API key.[cite: 1]
2. Open Postman, paste your API key, and send a test Kanglish word (like "yaru") to ensure the API returns "who" successfully.[cite: 1]
3. Once verified in Postman, use the AI prompt below to generate the Python function and integrate it into your main `ai_agent.py` file.[cite: 1]

AI Prompt: Bhashini Integration
"Act as an NLP Engineer.[cite: 1] Write a Python integration script for the Bhashini API that takes transliterated 'Kanglish' text, translates it into English for LLM processing, and then translates the LLM's English response back to Kanglish."[cite: 1]

Step 4: Deploy on Zoho Catalyst

Actionable Steps:[cite: 1]

1. Open your terminal and install the Zoho CLI by typing: `npm install -g zcatalyst-cli`.[cite: 1]
2. Type `catalyst login` in the terminal to connect your VS Code to your Zoho account.[cite: 1]
3. Upload your CSV files into the Catalyst Relational Data Store via their web dashboard.[cite: 1]
4. Ensure Docker Desktop is running on your laptop.[cite: 1]
5. Use the prompt below to generate your Dockerfile.[cite: 1] Use the Zoho CLI command `catalyst deploy` to push your Dockerized AI app to Catalyst AppSail.[cite: 1]

AI Prompt: Deployment
"Act as a DevOps Engineer.[cite: 1] Write a complete 'Dockerfile' for deploying a Python FastAPI Lang Chain app.[cite: 1] Include all dependencies like Pandas and LangChain.[cite: 1] Provide the terminal steps required to push this image to Zoho Catalyst AppSail."[cite: 1]

Kanglishnalli Vivaravada Mahiti (Deep Explanation in Kanglish)

Geleya, neevu kelidante ee project annu madalu bekaada apps mattu exact aagi enu madabeku annuva step-by-step vivarane illide:[cite: 1]

Bhaga 1: Install madabekada Apps

Neevu code bareyokke munche nimma laptop nalli ee apps irabeku:[cite: 1]

- **VS Code:** Illi neevu Python code type maduttira.[cite: 1]
- **Docker Desktop:** Zoho ge nimma app hakalu idu beku.[cite: 1] Idu laptop background nalli run agutirabeku.[cite: 1]
- **DB Browser:** Neevu srushtisida nakali data sariyagideya anta open madi nodokke.[cite: 1]
- **Postman:** Bhashini API sariyagi kelsa maduttideya anta code ge hako munche test madokke.[cite: 1]
- **Node.js:** Zoho CLI command galu nimma laptop nalli kelsa madalu idu beku.[cite: 1]

Bhaga 2: Step-by-Step Action Plan

Hanta 1: Data Srushti (Data Generation)
VS Code open madi faker mattu pandas install madi.[cite: 1] AI kotta code annu paste madi run madi.[cite: 1] Idu CSV files koduttade.[cite: 1] Aa file annu DB Browser nalli open madi, 1.5 laksha FIR mattu crime data KSP schema prakara sariyagideya anta check madi.[cite: 1]

Hanta 2: AI Medulu Nirmana (Agentic AI Core)
Ollama install madi `llama3` model pull madi. VS Code nalli LangChain mattu Ollama install madi. AI prompt balasi bandiruva code annu paste madi run madi.[cite: 1] Terminal nalli ondu prashne (ex: "highest accused iro CrimeNo yaru") anta type madi, AI sariyagi huduki uttara kodutideya anta test madi.[cite: 1]

Hanta 3: Kannada/Kanglish Alavadike
Bhashini website ge hogi API key togolli.[cite: 1] Modalu Postman app open madi, alli "yaru" anta type madi test madi, adu "who" anta translate agutideya nodi.[cite: 1] Sariyagiddare, AI kotta Python code annu nimma main code jote seri.[cite: 1]

Hanta 4: Zoho Catalyst nalli Hakuvudu (Deployment)
Terminal nalli `npm install` command balasi Zoho CLI install madi.[cite: 1] `catalyst login` madi nimma account connect madi.[cite: 1] Zoho website ge hogi nimma CSV data upload madi.[cite: 1] Docker Desktop open ittu, Dockerfile srushtisi, `catalyst deploy` command moolaka nimma project annu Zoho AppSail nalli host madi.[cite: 1]

All the best geleya, neevu idannu khandita gelluttira![cite: 1]